

NEMANIA BOROVITS

Researcher & AI Engineer

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List of Academic Publications:
🌐 <https://scholar.google.com/citations?user=VZOsmdAAAAAJ>

Professional skills

Data Analysis:



Programming languages: Python, SQL, C#.

Data engineering: Apache Spark, Apache Kafka, Elastic Stack, Microsoft SQL Server, MongoDB.

Containerized Execution: Docker, Kubernetes, Amazon ECS, Azure Kubernetes Service.

Data Visualization: Plotly Dash, Matplotlib, Seaborn, Kibana, Tableau, Power BI.

Machine learning: Scikit-learn, NumPy, Pandas, Snorkel.

Deep learning: TensorFlow, Keras, PyTorch.

NLP: NLTK, Hugging face Transformers, spaCy, gensim, t-SNE.

LLMs: RAG, LangGraph, MCP, Qdrant.

Web Crawling: BeautifulSoup.

Privacy Frameworks: Flower, Opacus, SEAL.

Certificates: CIPP/E (in progress).

Cloud Technologies:



Google Cloud

Heroku

Dataiku

Azure Databricks

Back-end Development:



Data modeling: Microsoft SQL Server, MongoDB.

ORM: Entity Framework, nHibernate.

Summary

I am a researcher and AI engineer with a background in software and data engineering. After five years in the industry in Greece, I relocated to the Netherlands to pursue an MSc in Data Science and Entrepreneurship, followed by a PhD in Data Engineering and AI for Privacy, co-financed by KPN. My expertise includes privacy engineering, privacy-preserving AI and data governance. With a decade of industry experience and five years in academia, I have demonstrated the ability to bridge research and practical applications. Currently, I am employed as a postdoctoral researcher on an EU-funded project supporting AI-driven decision-making for healthcare professionals addressing dementia and frailty. My passion is designing and developing contemporary software systems that generate impact in both industry and society.

Professional Experience

12/11/2024 Postdoctoral Researcher - System Architect

Tilburg University

I oversee the development and implementation of the COMFORTage platform, an AI-driven healthcare system for dementia and frailty, as part of a Horizon Europe-funded project involving 39 partners from academia and industry. My role includes designing the Reference Model to structure AI-assisted diagnostics and decision-making and specifying the Reference Architecture to standardize data processing, exchange and interoperability. I define system requirements to align technical implementations with ethical guidelines, data governance principles and interoperability standards. I design structured data models and governance mechanisms that support AI-assisted risk assessment and clinical decision-making, ensuring regulatory and operational compliance. I coordinate with technical and clinical stakeholders to integrate AI-enabled workflows into healthcare processes. I oversee the deployment of AI models within the platform, ensuring compliance with regulatory frameworks while supporting usability and adoption in clinical practice.

15/10/2020 - Data Scientist

11/11/2024 KPN

At the KPN Data Science Lab, I implemented Privacy by Design principles in data pipelines to enhance both analytics and governance processes.

Data Analytics: I collaborated with internal teams, external partners, and the legal department to ensure all projects were GDPR-compliant. My primary focus was on developing privacy-preserving NLP methods for analyzing Dutch call center text data. I worked closely with subject-matter experts to refine business KPIs and partnered with Data and MLOps engineers within an Agile/Scrum framework. Key projects included implementing rule-based, deep learning-based, and LLM-based data anonymization, designing privacy-preserving business ontologies and contributing to the development of novel privacy-compliant software for synthetic data evaluation from external vendors. I regularly provided project updates, created dashboards to support business and legal stakeholders, and presented findings to non-technical audiences.

Data Governance: To support KPN's transition to a decentralized data governance model, specifically a data mesh, I led the formulation and implementation of data contracts that incorporated privacy considerations from the outset, facilitating GDPR compliance monitoring. My responsibilities included drafting functional and non-functional requirements, policies, and data delivery agreements for the design and development of KPN's internal data products to serve all stakeholders effectively.

15/10/2020 - PhD Candidate

11/11/2024 Eindhoven University of Technology

Main Role: I was affiliated with the Jheronimus Academy of Data Engineering Lab (JADE Lab). As a PhD candidate, I conducted research on privacy engineering and privacy-preserving AI at the Dutch telecommunications provider KPN. I contributed to several Horizon Europe-funded projects, including SODALITE H2020 and RADON H2020 (Infrastructure as Code), MARIT-D (EU maritime trafficking intelligence), and

Business Logic: asp .net MVC, asp .net (C#).

API: fastAPI, Flask, asp .net Web API (C#), .

Web Services: SOAP, RESTful (C#, Python).

Front-end development:



Web skills: HTML5, Javascript, jQuery, AJAX, JSON, CSS, Bootstrap.

JS Frameworks: React js, Knockoutjs, VueJs.

Source control tools:



GitHub, BitBucket.

COMFORTage (AI models for dementia and frailty prevention in healthcare). My research led to publications in Q1 journals and presentations at rank-A conferences in privacy engineering, privacy-preserving AI, software engineering and infrastructure-as-code.

Education: I taught three master 's-level courses at JADS in the Data Science and Business Entrepreneurship program: Data Forensics, Natural Language Processing (NLP) and Data Mining. I also co-supervised 12 Master's students on theses in privacy engineering, privacy-preserving AI, and text analytics. Community Engagement: I have actively participated in the JADS PhD Board since October 2022. I organized community-building activities for PhD candidates and delivered public talks about my research at JADS events.

15/10/2019 - Data Scientist / Research Assistant
31/03/2020 Tilburg University

01/10/2016 - Data Engineer
17/08/2018 Mouzenidis Group

01/10/2014 - Full Stack Developer
30/09/2016 Mouzenidis Group

10/04/2013 - Junior Web Developer
30/09/2014 Mouzenidis Travel

Education

15/10/2020 - PhD in Data Engineering and AI for Privacy
23/06/2026 Eindhoven University of Technology

27/08/2018 - MSc Data Science & Entrepreneurship
10/07/2020 Eindhoven University of Technology

01/10/2006 - BSc in Applied Informatics
18/07/2012 University of Macedonia

Languages

Mother tongues:

Greek, Serbian.

Other languages:

English: IELTS, Proficiency in English (University of Michigan).

Dutch: Basic communication skills / working knowledge.

Organizational Competencies

Creating and keeping deadlines	Project management	Teamwork
Goal setting and meeting goals	Creative thinking	Strategic thinking

Other interests

Artificial intelligence ethics enthusiast.

Jogging, playing football and basketball, cooking and travelling.